

RadiomiXAI: A Radiomics Foundation Model for Addressing Feature Importance Confounding

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Abstract. Radiomics features are often highly intercorrelated because of latent factors such as tumor volume, which can distort feature importance estimates in machine learning models and limit interpretability. We present RadiomiXAI, a model-agnostic explainability framework built on a radiomics foundation model trained on large-scale Open-radiomics data to capture reusable feature dependency structure and identify volume-related confounding. A total of 476,424 radiomics feature vectors were extracted with PyRadiomics, and correlation analysis identified 74 features strongly associated with tumor volume. Standard normalization methods, including z-score scaling, min-max scaling, log transformation, and volume normalization, were evaluated for their ability to reduce multicollinearity. A Bayesian-network-based foundation model was then trained on radiomics features to learn global dependencies. The model was used to derive uncertainty bounds for perturbation-based feature importance estimates from a LightGBM classifier for glioma grading on contrast-enhanced T1-weighted magnetic resonance imaging (MRI). We showed that conventional normalization methods were ineffective in ad-

addressing the confounding effect of tumor volume, with only 1 of 74 features improving after min-max scaling and 4 after volume normalization. Although the baseline importance estimate for tumor volume was low at 0.0087, dependency-aware analysis revealed a substantially higher upper bound of 0.2059, indicating that tumor volume acts as a latent confounder influencing multiple downstream features. The findings show that standard normalization is insufficient and that dependency-aware adjustment can improve the interpretability and reliability of radiomics models.

Keywords: Radiomics · Explainable Artificial Intelligence · Foundation Models · Feature Importance · Glioma Grading

1 Introduction

Radiomics converts routine clinical images into quantitative descriptors of shape, intensity, and texture, providing a noninvasive route to imaging biomarkers and model-based decision support in oncology [12, 8]. Across cancer imaging, radiomic signatures have been explored for prognosis, risk stratification, treatment response assessment, and phenotype characterization [2, 22, 23]. Because these features can be derived from standard-of-care imaging, radiomics fits naturally within precision medicine workflows and continues to attract interest as a scalable companion to clinical and molecular data [24].

Clinical translation, however, remains limited [14, 4]. Reported gains often depend on scanner settings, segmentation choices, preprocessing details, and software implementations, which weakens reproducibility and external validity [5]. As a result, predictive performance alone is rarely sufficient for clinical adoption; models must also produce explanations that clinicians can trust.

Feature importance in radiomics is therefore crucial. It is frequently used to identify candidate biomarkers, explain model behavior, and justify downstream biological interpretation. Yet this practice assumes that predictors contribute in a sufficiently separable way. In radiomics, that assumption is weak as features are highly correlated, often organized by shared latent structure, and may encode overlapping properties of the same lesion.

Tumor volume is one of the clearest sources of this shared structure. Many texture- and intensity-derived variables are directly or indirectly influenced by lesion size. Thus, apparent importance can propagate across the feature space rather than remain localized to a single variable. In that setting, a model may appear to rely on diverse imaging descriptors while partly reflecting volume-related effects, which complicates biological interpretation.

Standard preprocessing offers only partial relief. Scaling, filtering, and multicollinearity checks such as the variance inflation factor (VIF) can highlight redundancy. However, they do not model how dependencies propagate across the feature space [19]. The same limitation affects common explainability tools. Permutation-based importance and SHapley Additive exPlanations (SHAP)-style methods are widely used [20, 3, 13], but their assumptions weaken when

features are strongly dependent [9, 1, 18, 15]. Under these conditions, explanations can become unstable or misleading [7].

We therefore treat radiomics as a structured system rather than a set of independent predictors. In this work, we introduce RadiomiXAI, a model-agnostic framework that uses a Bayesian-network-based radiomics foundation model to capture global dependencies and reinterpret feature importance under structured correlation. We evaluate the approach on large-scale radiomics data and a glioma grading task, with the goal of producing importance estimates that are more reliable for biomarker discovery and clinical interpretation.

2 Methodology

To mitigate the NP-hard computational complexity inherent in large-scale Bayesian network structure learning, RadiomiXAI was explicitly designed as a reusable foundation model, decoupling the resource-intensive global dependency mapping from the computationally lighter task-specific model evaluation. We first extracted radiomics feature vectors from Open-radiomics [16] and included tumor volume as an explicit variable in the analysis pipeline. We then identified volume-associated features, evaluated whether conventional normalization methods reduced multicollinearity, and learned a global dependency structure using a Bayesian network. Separately, we trained a LightGBM classifier for glioma grading on contrast-enhanced T1-weighted MRI and estimated baseline feature importance using perturbation analysis on the validation set. Finally, we used the learned dependency structure to interpret and adjust feature importance estimates under structured feature dependence.

2.1 Dataset

Open-radiomics served as the source dataset for both large-scale dependency learning and task-specific model evaluation [16]. Across the available collections, a total of 476,424 radiomics feature vectors were extracted using PyRadiomics. For the glioma grading experiment, we used contrast-enhanced T1-weighted MRI and defined the enhancing tumor core as the region of interest (ROI) for feature extraction. The predictive task was binary classification of high-grade versus low-grade glioma. Tumor volume was retained as an explicit variable alongside the radiomics features so that its direct and propagated effects could be quantified during subsequent analysis. Apart from the normalization strategies evaluated in the statistical analysis stage, no additional aggressive preprocessing was introduced beyond the standard feature extraction workflow. The disease setting is consistent with prior pediatric and glioma-focused radiomics studies [17, 6, 22].

2.2 Statistical Analysis

The statistical analysis stage was designed to characterize the dependency structure of the radiomics feature space before task-specific model interpretation. This

stage included correlation analysis, multicollinearity assessment under common normalization schemes, Bayesian network learning, and dependency-aware adjustment of feature importance.

Correlation Analysis We computed the Pearson correlation coefficient between each radiomics feature and tumor volume to identify strong direct linear associations with lesion size. Features with correlation magnitude greater than 0.8 were classified as strongly volume-associated. This analysis provided a baseline characterization of potential confounding and identified the subset of features most directly linked to tumor volume.

Multicollinearity Assessment To evaluate whether standard preprocessing methods reduced dependency within the radiomics space, we assessed multicollinearity using VIF [19]. We compared raw features with features processed using z-score normalization, min-max scaling, log transformation, and volume normalization. A VIF greater than 10 was treated as evidence of substantial multicollinearity. This analysis was used to determine whether conventional normalization strategies meaningfully reduced dependence on tumor volume and related feature redundancy.

Bayesian Network for Dependency Modeling To model global dependency structure, we constructed a Bayesian network that treated radiomics features and tumor volume as nodes in a probabilistic graphical model [10]. In this representation, learned edges corresponded to conditional dependencies among variables rather than simple pairwise associations. Network structure was learned using a score-based procedure applied to the large-scale feature set, and graphics processing unit (GPU)-accelerated data processing was used to support efficient computation at scale. The resulting graph enabled analysis of both incoming and outgoing dependency patterns for tumor volume and other features, allowing us to identify how influence could propagate across the radiomics space through indirect pathways. Unlike pairwise correlation analysis, this framework captured global structure and conditional dependence relationships across the full feature system.

Dependency-Aware Importance Adjustment The learned Bayesian network was then used as the foundation of a dependency-aware interpretation framework. Starting from baseline feature importance estimates obtained from the predictive model, we used the network structure to evaluate how importance could be redistributed through dependency pathways rather than remaining localized to a single variable. This procedure yielded uncertainty-aware bounds on feature importance and allowed us to distinguish directly predictive features from features whose apparent importance was amplified by correlated or propagated effects. The goal was not to replace baseline importance scores, but to contextualize them within the global structure of the radiomics feature space. All codes and models will be available on our GitHub repository upon acceptance.

2.3 Machine Learning

For task-specific evaluation, we trained a LightGBM classifier for binary glioma grading using radiomics features extracted from the enhancing tumor core on contrast-enhanced T1-weighted MRI [11]. The dataset was partitioned into training, validation, and test sets using a random 60/20/20 split. All model development decisions, including feature importance estimation, were restricted to the training and validation stages to avoid information leakage from the held-out test set. The classifier was trained on the extracted radiomics features, and feature importance was estimated on the validation set using a perturbation-based procedure that measured the mean absolute change in predicted probability after feature perturbation. These values served as baseline importance estimates. We then compared baseline and dependency-adjusted importance values to determine how the learned Bayesian network modified interpretation under structured feature dependence.

2.4 Software and Hardware

All experiments were conducted in Python 3.12.7 on a workstation running Ubuntu 24.04.2 LTS with CUDA 12.8 and cuDNN enabled. The software stack included PyRadiomics for feature extraction, LightGBM for classification, and standard scientific Python libraries including NumPy, SciPy, and pandas for data processing and analysis. GPU-accelerated processing with cuDF was used for large-scale feature handling, and the Bayesian network component was implemented using a Python probabilistic graphical modeling library. The workstation was equipped with an AMD Ryzen Threadripper 2990WX processor, 128 GB of RAM, and an NVIDIA GeForce RTX 5090 GPU.

3 Results and Discussion

Correlation analysis identified 74 radiomics features that were strongly associated with tumor volume, using a Pearson correlation threshold of $r > 0.8$. This finding confirmed the presence of a dominant dependency structure within the radiomics feature space. Variance inflation factor analysis showed that this dependency persisted across the identified features and was not meaningfully reduced by conventional preprocessing. After z-score scaling, all 74 features remained highly multicollinear. The same was true after logarithmic transformation, with all 74 features still exceeding the multicollinearity threshold. Min-max scaling reduced the count by only one, leaving 73 of 74 features highly multicollinear, while volume normalization reduced the count to 70 of 74. These results indicate that the observed dependence is not a superficial artifact that can be removed through routine preprocessing.

The Bayesian network provided a more detailed view of this structure and placed tumor volume, represented by *original_shape_VoxelVolume*, at the center of a volume-centric dependency graph. As shown in Fig. 1, tumor volume acted

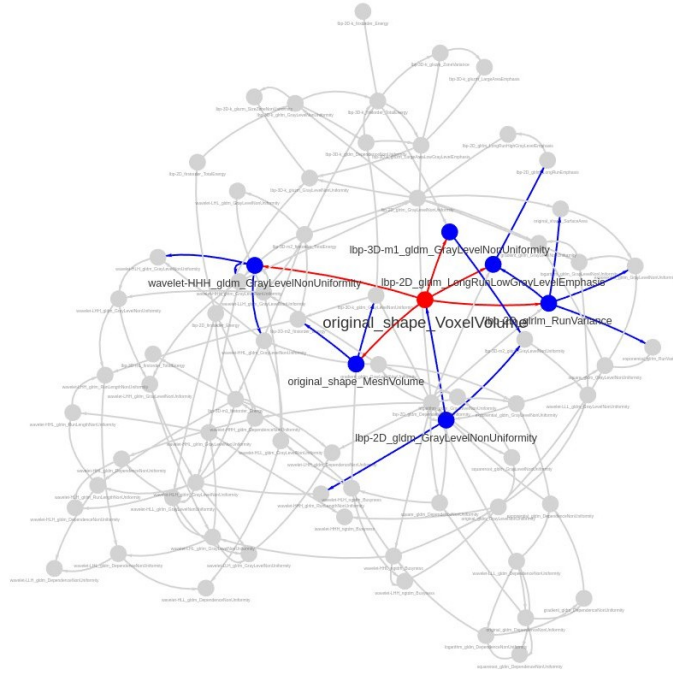


Fig. 1. Bayesian network learned from radiomics features and tumor volume.

as a hub node with multiple outgoing connections to downstream radiomics variables. Many of these dependent variables were texture- and intensity-derived features that are often interpreted as biologically informative descriptors. The network topology therefore suggests that tumor volume influences the radiomics space indirectly through propagated dependencies, rather than only through a direct association with the classification target.

Baseline perturbation-based importance analysis with the LightGBM classifier produced a ranking dominated by several texture- and shape-derived features. The top baseline features included *LongRunEmphasis* from *lbp-2D_gldm* (0.0961) and *DependenceNonUniformity* from *wavelet-LLH_gldm* (0.0873). Other highly ranked variables were the original-shape feature *SurfaceArea* (0.0745), *GrayLevelNonUniformity* from *lbp-3D-k_glszm* (0.0719), and *LargeAreaLowGrayLevelEmphasis* from *lbp-3D-k_glszm* (0.0702). In contrast, tumor volume itself showed a low baseline importance of 0.0087, which under a conventional interpretation would suggest that it played only a minor role in prediction. The baseline ranking is shown in Fig. 2. On its own, this result could be taken to mean that the classifier was driven primarily by non-volume texture descriptors. However, that interpretation depends on the assumption

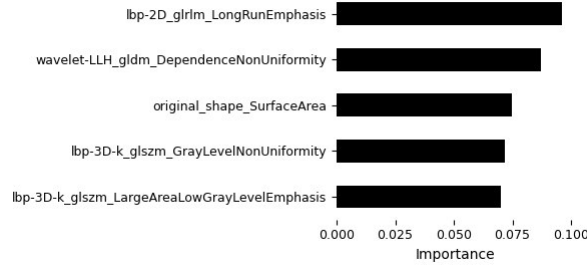


Fig. 2. Baseline perturbation-based feature importance estimates from the LightGBM classifier.

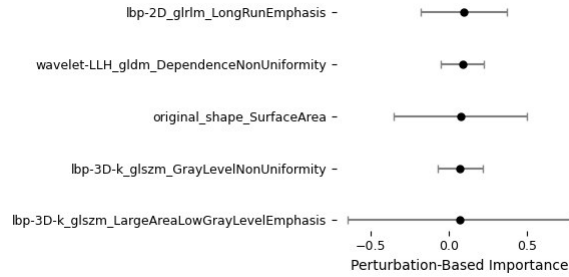


Fig. 3. Dependency-aware feature importance estimates after incorporation of the Bayesian network.

that perturbing one feature isolates its contribution from the rest of the feature space.

When the Bayesian network was incorporated into the interpretation stage through RadiomiXAI, the importance landscape changed substantially. Tumor volume showed the clearest example of dependency-driven underestimation in the baseline analysis. Its baseline importance of 0.0087 increased to an upper bound of 0.2059 after accounting for outgoing dependency pathways, indicating a much larger indirect influence on prediction than was visible under conventional feature perturbation. Several other features also exhibited broad uncertainty intervals, consistent with instability induced by shared dependencies. For example, *Energy* from *lbp-3D-m1_firstorder* spanned $[-0.037, 0.938]$, whereas *LargeAreaLowGrayLevelEmphasis* from *lbp-3D-k_glszm* spanned $[-0.159, 1.267]$. In the same analysis, the original-shape feature *SurfaceArea* spanned $[0.075, 0.922]$, and *LongRunEmphasis* from *lbp-2D_glrIm* spanned $[0.096, 0.643]$. These widened intervals indicate that the apparent importance of highly ranked features was not fully intrinsic to those variables, but depended in part on how influence propagated through the surrounding feature network.

A complementary pattern was seen among features with low or zero baseline importance. Features such as *GrayLevelNonUniformity* from *lbp-2D_gldm* and the corresponding *GrayLevelNonUniformity* term from *original_gldm* had intervals of $[-0.049, 0.000]$, indicating that their apparent contribution was likely attributable to upstream dependencies rather than to independent predictive value. This is an important distinction because conventional rankings tend to flatten these relationships into a single ordered list, even when multiple features are partially expressing the same underlying signal. In this setting, the revised importance estimates do not simply reorder features. They expose which features are stable contributors, which are dependency-sensitive, and which are likely inheriting relevance from more central variables. The framework also permits closely related volume descriptors to remain in the analysis simultaneously; for example, *original_shape_VoxelVolume* and *original_shape_MeshVolume* need not be reduced to an arbitrary either-or choice.

These findings clarify why conventional normalization and explainability methods are insufficient in this setting, consistent with recent analyses of volume confounding and translational fragility in radiomics [21, 14, 4]. The failure of pre-processing to remove high multicollinearity indicates that the radiomics space is organized by persistent structural dependencies. The Bayesian network further shows that tumor volume is embedded within this structure as a major source of propagated influence. Comparing baseline and dependency-aware importance estimates therefore suggests that standard feature ranking can obscure that role. More broadly, some features that appear biologically informative under standard analysis may instead be downstream expressions of size-related effects.

From a scientific and clinical perspective, these results imply that radiomics-based biomarker discovery should account for dependency structure before mechanistic importance is assigned to individual features. In this study, dependency-aware importance estimation consistently showed stronger influence from tumor volume and volume-associated pathways than baseline methods suggested, while many texture features exhibited inflated or unstable importance because of shared structure.

4 Conclusion

RadiomiXAI demonstrates that radiomics feature importance can be substantially distorted by hidden dependency structures, with tumor volume acting as a central latent driver that propagates influence across multiple features. Conventional normalization and standard explainability methods fail to account for this effect, leading to potentially misleading interpretations. By explicitly modeling feature dependencies through a Bayesian network, RadiomiXAI provides uncertainty-aware, dependency-adjusted importance estimates that more accurately reflect true feature contributions. This framework highlights the need to move beyond independent feature assumptions in radiomics and supports more reliable biomarker discovery and clinically interpretable artificial intelligence models.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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